

Heredity and Evolution

I] Textbook Questions:

1 Mark Objectives:

1. Complete the statements by choosing correct options from bracket.

(Gene, Mutation, Translocation, Transcription, Gradual development, Appendix)

- a. The causality behind the sudden changes was understood due to **mutation** principle of Hugo de Vries. [2026]
- b. The proof for the fact that protein synthesis occurs through **gene** was given by George Beadle and Edward Tatum.
- c. Transfer of information from molecule of DNA to mRNA is called as **transcription** process. [2020]
- d. Evolution means **gradual development**. [2024]
- e. Vestigial organ **appendix** present in human body is proof of evolution. [2023]

2 Marks Questions:

Q. Write short notes:

1. Lamarckism. Or

Explain the theory of Lamarckism with the help of any two suitable examples. [2026]

Ans:

1. Lamarck proposed that morphological changes occurring in living organisms are responsible for evolution and the reason behind these morphological changes is activeness or laziness in that organism.
2. For example, neck of the giraffe has become too long due to browsing on leaves of tall plants by extending their neck for several generations.
3. Shoulders of ironsmith have become very strong due to frequent hammering movements.
4. He called this concept as principle of use or disuse of organs.
5. He proposed that living organisms can transfer the characters which they have acquired during their lifetime to the next generation.
6. This is called theory of inheritance of acquired characters.

2. Darwin's theory of natural selection.

[2020, 2023]

Ans:

1. Darwin proposed the theory of natural selection which explains the survival of the fittest.
2. According to him, all the organisms reproduce prolifically and compete in a life-threatening manner.
3. Only those organisms sustain which show the modifications essential for winning the competition. Nature selects only those organisms which are fit to live and rest perish.
4. These sustained and selected organisms reproduce and thereby give rise to new species with their own specific characters.

3. Embryology.

Or There are evidences of foetal science among chordates.

Ans:

1. Study of the development of embryos is called embryology.
2. Embryology of various vertebrates shows that all embryos show extreme similarities during initial stages and those similarities decrease gradually.
3. Embryos of fish, salamander, tortoise, chicken, pig, cow, rabbit, and man look similar in initial stages.
4. Embryology is used as one of the pieces of evidence of evolution as similarity in initial stages of development indicates common origin of all the animals.

4. Evolution

Ans.

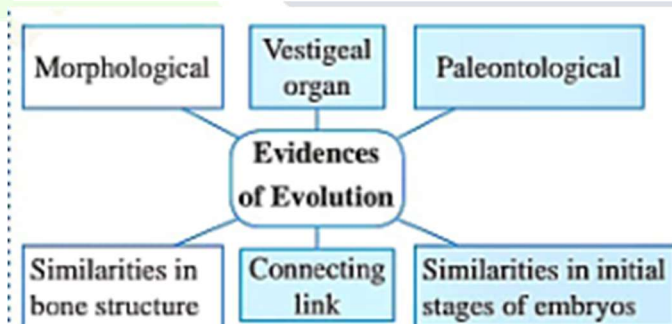
1. Evolution is the gradual change occurring in living organisms over a long duration.
2. This is a very slow process through which development of organisms is achieved.
3. All the stages in changes which occurred in various components, ranging from stars and planets in space to the biosphere present on Earth, are included in evolution.
4. Formation of new species due to changes in specific characters of several generations of living organisms as a response to natural selection is called evolution.

5. Connecting links OR Explain the concept of connecting links with two examples. [2022]

Ans.

1. Some plants and animals show some morphological characters by which they can be related to two different groups; hence they are called connecting links.
2. Connecting links prove that organisms have evolved.
3. For example – In Peripatus, characters like segmented body, thin cuticle, and parapodia-like organs are present. Similarly, these animals show tracheal respiration and open circulatory system like arthropods.
4. Hence, this indicates that Peripatus is a connecting link between Annelida and Arthropoda.
5. Lungfish performs respiration with lungs (like amphibians) but it is a fish. These examples indicate that mammals have evolved from reptiles, and amphibians from fish.

6. Complete the flow chart: [2019,2021, 2022]



3 Marks Questions:

1. Study of fossils is an important aspect of study of evolution.

OR

What do you mean by fossils? Write the importance of fossils. Write the name of method to find the age of fossils.

[2021]

OR Define fossils. Explain the importance of fossils as proof of evolution.

[2021]

Ans.

1. Many organisms get buried due to disasters like earthquake, flood, and volcano.
2. Remnants and impressions of these organisms remain preserved underground. These are called fossils.
3. Fossils of organisms that existed millions of years ago could be studied. Carbon dating method is used to determine the age of fossils.
4. Once the age of the fossil has been determined, it becomes possible to deduce information about the organism and other erstwhile organisms.
5. The deepest fossils were those of invertebrates, above them were those of fishes, amphibians, and reptiles, followed by those of mammals and aves.
6. This shows that vertebrates have slowly evolved from invertebrates. Thus, study of fossils is an important aspect of study of evolution.

Q. Answer the following questions.

2. Explain with suitable examples importance of anatomical evidences in evolution.

[2019, 2020]

Ans:

1. The study of the internal structure of organisms is called anatomy.
2. Anatomical similarities help as evidence in the theory of evolution.
3. A human hand, forelegs of ox, patagium of bat, or flipper of a whale. The use of each of these structures in different animals is different.
4. e.g. a human hand is used to pick up things, forelegs of ox are used to walk, bat's patagium is used to fly, and flippers of whales are used to swim.
5. However, there is similarity in the structure of bones and bone-joints in the above-mentioned organs of each of these animals.
6. This similarity indicates that these animals may have a common ancestor. Thus, anatomical evidence helps in understanding the evolution process.

3. Define heredity. Explain the mechanism of hereditary changes.

[2025]

Ans:

1. Heredity is the transfer of biological characters from one generation to another through genes.
Mechanism of hereditary changes:
2. Living organisms can produce new individuals like themselves due to genes. Some genes are transmitted from one generation to the next without changes.
3. Due to this, some characters of parents are transmitted to their offsprings. However, sometimes sudden changes occur in those genes.
4. Sometimes, a nucleotide of the gene changes its position that causes a minor change, which is called mutation.
5. Some mutations may be minor but some may be considerable, and may also cause genetic disorders like sickle cell anaemia.

4. Define vestigial organs. Write the names of some vestigial organs in human body and write the names of those animals in whom the same organs are functional.

[2020, 2024, 2025]

OR Define vestigial organs. Write names of any two vestigial organs in human body. [2019]

Ans:

1. Degenerated or underdeveloped useless organs of organisms are called vestigial organs.
2. Some vestigial organs in human body which are functional in other organisms are as follows:
 - a. Appendix – functional in ruminants
 - b. Muscles of pinna – functional in monkeys for movement of pinna
 - c. Tailbone (coccyx) – functional in arboreal (tree living) animals and monkeys
 - d. Body hair – functional in some animals to protect the body and maintain body temperature

5 Marks Questions:

5. Explain the process of formation of complex proteins.

**This question is never asked completely for 5 marks but selected topics are asked from it in the exams for 1 to 3 marks.*

Ans:

Information about protein synthesis is stored in DNA. These proteins are synthesized by DNA through the RNA. This is called as '**Central Dogma**'

It is carried out in three steps:

1. Transcription:

1. mRNA is produced as per the sequence (complimentary) to the nucleotides on DNA.
2. This process of RNA synthesis is called as transcription.

2. Translation:

1. The mRNA formed in the nucleus comes in the cytoplasm. It brings in a coded message from the DNA. The message contains codes for amino acids.
2. The code for each amino acid consists of three nucleotides. It is called triplet codon.
3. As per the message on mRNA, amino acids are supplied by tRNA. For this purpose, tRNA has anticodon. This is called translation.

3. Translocation:

1. The amino acids brought in by tRNA are bonded together by peptide bonds with the help of rRNA.
2. During this process, the ribosome keeps on moving from one end of the mRNA to the other. This is called translocation.
3. Many such chains come together to form complex proteins.

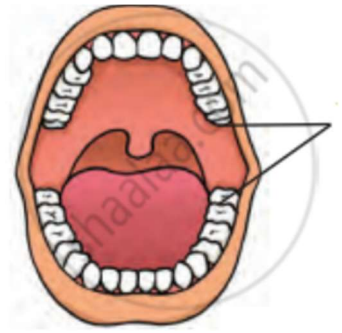
II] Extra - Previous Years' Questions: (PYQs)

1 Mark Objectives:

1. Any nucleotide of the gene suddenly changes its position that causes a minor change which is nothing but the **mutation**. [2025]
(A) changes (B) mutation (C) disability (D) translocation
2. Genetic disorders like sickle cell anaemia may be caused due to **mutation**. [2023]
(A) mutation (B) translation (C) translocation (D) transcription
3. Human males have **44 + XY** numbers of chromosomes. [2023]
(A) 44 + XX (B) 44 + XY (C) 22 + XY (D) 22 + XX

Q. True or False:

4. Evolution is gradual change occurring in living organisms over a long duration. (True) [2025]
5. Study of fossils is an important aspect of the study of evolution. (True) [2025]
6. write the name of the indicated part. **Ans :** Wisdom teeth [2023]
2. write the scientific name of human being. **Ans:** Homo sapiens [2022]



2 Marks Questions:

1. Observe the given animals and answer the questions: [2025]



Fig. A

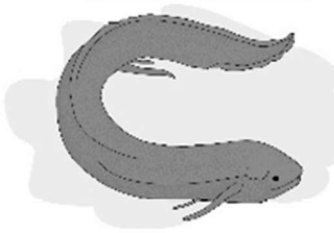


Fig. B

- a. Identify the animals shown. **Ans:** Fig. A: Duckbilled platypus Fig. B: Lung fish
- b. State the type of evidence of evolution related to above animals. **Ans:** Connecting links

2. Give Reason: Duck billed platypus is a connecting link. [2023]

Ans:

1. Some plants and animals show some morphological characters by which they can be related to two different groups; hence they are called connecting links.
2. Duck billed platypus is a connecting link between reptiles and mammals.
3. Duck billed platypus lays eggs like reptiles.
4. But it also shows presence of mammary glands and hairs like the mammals.

3. Write any two objections that were raised against Darwin's Theory of Natural selection: [2022, 2023]

Ans:

Following is some of the main objections raised against Darwin's theory of natural selection:

- a. Natural selection is not the only factor responsible for evolution.
- b. Darwin did not mention any explanation about useful and useless modifications.
- c. There is no explanation about slow changes and abrupt changes.

4. What is meant by carbon dating? Give any one use of this technique. [2026]

Ans:

1. Carbon dating is a method used in determining the age of fossils and manuscripts
2. The time passed since the death of a plant or animal can be calculated by measuring the radioactivity of C-14 and the ratio of C - 14 and C - 12 present in their body.
3. Once the age of fossil has been determined, it becomes easy to deduce information about other erstwhile organisms.
4. It seems that vertebrates have been slowly originated from invertebrates.